MIL-T-13333 (Ord)
23 March 1954
SUPERSEDING
FXS-647
10 Jan 1945

MILITARY SPECIFICATION

TELESCOPES, OBSERVATION, M48 AND M49

1. SCOPE

1.1 This specification covers two models of observation telescopes used for observing affectiveness of artillery fire. These two telescopes are almost identical optically, but differ slightly in their construction.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS AND PUBLICATIONS

2.1 The following specifications and drawings, and specifications and drawings referenced thereon, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Military

MIL-P-13123	Packaging of Telescopes, Observation, M48 and M49

U. S. Army

51-7	Instruments, Fire-Control; General Specification Governing the Manufacture, and Inspection of
51-70-2	Optical Components for Fire-Control Instruments; General Specification Governing the Manufacture, Assembly and Inspection of

DRAWINGS

Ordnance Corps

Class 22, Division 345	Telescope, Observation, M48
Class 22, Division 346	Telescope, Observation, M49

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(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions ;should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- General requirements and associated tests specified in Specifications 51-7 and 51-70-2 which are common to all fire control instruments shall be applicable to this specification.

3.2 Materials.- Materials shall be in accordance with the applicable drawings, and specifications specified herein and referenced on drawings, Class 22, Division 345 and Class 22, Division 346.

3.3 Accuracy.-

3.3.1 Definition.- The resolving power at the geometric axis for a parallel bar test target shall be 3-1/2 seconds of arc or better.

3.3.2 Eyepiece.- Eyepiece shall focus on objects from 50 feet to an indefinite distance from the observer.

3.3.3 Image tilt.- The image of a plumb line shall be erect within 45 minutes of arc (equivalent to 0.3 mil block measured across the field of view) as measured by a collimating telescope placed in front of the eyepiece.

3.4n Temperature.- The temperature requirements and test Type I contained in Specification 51-7 shall apply except minus 40°F shall be substituted for minus 45°F, and minus 80°F substituted for minus 60°F.

3.5 Vibration.- The telescopes shall be vibrated between 2 and 2-1/2 minutes at an amplitude of 1/16 inch (1/8 inch total displacement) and a frequency of 30 cycles per second, after which the telescope shall not contain loose dirt in excess of the following:

3.5.1 Dirt on the inner surface of the objective lens and the adjacent surface of the Porro Prism shall not exceed four in number (two on each surface) of a size not larger than a 50 dig.

3.5.2 Dirt o the adjacent faces and exit face of the Porro Prism shall not exceed three in number of a size not larger than a 50 dig.

3.5.3 Dirt on both faces of the Field Lens and adjacent surface of the eye lens shall not exceed six in number (two on each face) of a size not larger than a 20 dig.

3.6 Workmanship.- Workmanship shall be in accordance with Specifications 51-7 and 51-70-2.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection.- Unless otherwise specified, each telescope shall be inspected for conformance with this specification and as specified in Specifications 51-7 and 51-70-2.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Specification MIL-P-13123.

6. NOTES

6.1 Ordering data.- Invitation for bids and contracts should specify the title, number, date of this specification, and type of packing.

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